Mathematical Foundations of N-of-1 Trial Simulation: A Comprehensive Analysis

pmsimstats2025 Project Documentation

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Abstract

This document provides an in-depth, comprehensive synthesis of all mathematical elements underlying the `pmsimstats2025` N-of-1 clinical trial simulation framework. The analysis draws from extensive documentation and represents a unified treatment of the project's theoretical foundations, covering covariance matrix architecture, positive definiteness constraints, correlation structures, carryover modeling, and mixed-effects model specification.

Contents

Part I

Covariance Matrix Architecture

1 The Core Identity: $\Sigma = DRD$

The fundamental mathematical identity governing the simulation is the decomposition of the covariance matrix into correlation and variance components:

$$\boxed{\Sigma = D \cdot R \cdot D} \tag{1}$$

where:

- Σ is the full covariance matrix (26×26 for 8-timepoint designs)
- R is the correlation matrix (dimensionless, all diagonal entries = 1)
- D is a diagonal matrix of standard deviations

This decomposition separates two distinct concerns:

1. **Correlation structure (R)**: How variables relate to each other (independent of scale)
2. **Variance magnitude (D)**: The scale of each variable's variability

2 Block Partitioning Strategy

The full covariance matrix is partitioned into blocks for computational efficiency:

$$\Sigma = \begin{bmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{12}' & \Sigma_{22} \end{bmatrix} \tag{2}$$

Table 1: Block Matrix Components

Block	**Dimension**	**Contents**
Σ_{11}	$(3n_{tp}) \times (3n_{tp})$	Response components (BR, ER, TR) at all timepoints
Σ_{22}	2×2	Biomarker and baseline
Σ_{12}	$(3n_{tp}) \times 2$	Cross-covariances

For 8-timepoint designs: Σ_{11} is 24×24, giving a full matrix of 26×26.

3 Intra-Block Structure

Within Σ_{11}, there is further structure:

$$\Sigma_{11} = \begin{bmatrix} \Sigma_{BR} & C_{BR,ER} & C_{BR,TR} \\ C_{ER,BR} & \Sigma_{ER} & C_{ER,TR} \\ C_{TR,BR} & C_{TR,ER} & \Sigma_{TR} \end{bmatrix} \tag{3}$$

Each diagonal block (e.g., Σ_{BR}) is an $n_{tp} \times n_{tp}$ matrix with:

- Diagonal entries = 1 (unit variance for correlation matrix)
- Off-diagonal entries = autocorrelation parameter (e.g., $c_{br} = 0.75$)

Cross-blocks (e.g., $C_{BR,ER}$) contain:

- Diagonal entries = c_{cf1t} (same-timepoint cross-correlation)
- Off-diagonal entries = c_{cfct} (different-timepoint cross-correlation)

Part II
Conditional Multivariate Normal Sampling

4 The Two-Stage Algorithm

Rather than inverting the full 26×26 matrix, the simulation uses the Schur complement to enable efficient two-stage sampling:

4.1 Stage 1: Generate Baseline Variables

Generate (biomarker, baseline) from Σ_{22}:

$$(x_2) \sim \mathcal{N}(\mu_2, \Sigma_{22}) \tag{4}$$

4.2 Stage 2: Generate Response Components

Generate response components conditional on Stage 1:

$$x_1|x_2 \sim \mathcal{N}(\mu_{1|2}, \Sigma_{cond}) \tag{5}$$

where:

$$\mu_{1|2} = \mu_1 + \Sigma_{12}\Sigma_{22}^{-1}(x_2 - \mu_2) \tag{6}$$

$$\Sigma_{cond} = \Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{12}' \tag{7}$$

5 Computational Advantages

This approach reduces computational complexity significantly:

Table 2: Computational Comparison

Operation	**Full Matrix**	**Two-Stage**
Matrix inversion	26×26	2×2
Cholesky decomposition	$O(26^3)$	$O(2^3) + O(24^3)$
Numerical stability	Lower	Higher

The key insight is that Σ_{22} is only 2×2, making its inversion trivial and numerically stable.

6 Conditional Covariance Requirements

The conditional covariance Σ_{cond} is the central matrix for data generation. It must be:

1. **Positive definite** (all eigenvalues > 0)
2. **Well-conditioned** (condition number $\kappa < 100$)
3. **Numerically stable** for Cholesky decomposition

Part III
Positive Definiteness Constraints

7 Mathematical Definition

Definition 1 (Positive Definiteness)**.** A symmetric matrix Σ is positive definite if and only if:

$$\mathbf{v}^T \Sigma \mathbf{v} > 0 \quad \forall \mathbf{v} \neq \mathbf{0} \tag{8}$$

Equivalently: All eigenvalues $\lambda_i > 0$.

8 Verification Methods

Three methods are used to verify positive definiteness:

8.1 Method 1: Sylvester's Criterion

Theorem 1 (Sylvester's Criterion)**.** A symmetric matrix is positive definite if and only if all leading principal minors are positive:

$$\det(\Sigma_k) > 0 \quad \text{for } k = 1, 2, \ldots, n \tag{9}$$

where Σ_k is the $k \times k$ upper-left submatrix.

8.2 Method 2: Eigenvalue Criterion

$$\lambda_{min}(\Sigma) > 0 \tag{10}$$

This is the primary method used in `build_sigma_matrix()`.

8.3 Method 3: Cholesky Decomposition

If $\Sigma = LL^T$ exists (with L lower triangular, positive diagonal), then Σ is PD.

9 Gershgorin Circle Theorem

Theorem 2 (Gershgorin)**.** For eigenvalue bounds:

$$\lambda_i \in \bigcup_{j=1}^{n} \{z \in \mathbb{C} : |z - a_{jj}| \leq R_j\} \tag{11}$$

where $R_j = \sum_{k \neq j} |a_{jk}|$ is the off-diagonal row sum.

For correlation matrices (diagonal = 1):

$$\lambda_i \in [1 - R_j, 1 + R_j] \tag{12}$$

This implies that if off-diagonal sums exceed 1, negative eigenvalues become possible.

10 The Correlation Hierarchy Constraint

The critical constraint for PD preservation is the correlation hierarchy:

$$\boxed{c_{cfct} < c_{cf1t} < \min(c_{tv}, c_{pb}, c_{br})} \tag{13}$$

Remark 1. Measurements at different times cannot be more correlated than measurements at the same time. Violating this creates impossible constraint systems.

Hendrickson's values satisfy this:

- $c_{cfct} = 0.05 < c_{cf1t} = 0.12 < c_{br} = 0.75$ ✓

11 Biomarker Correlation Bound

Empirical analysis establishes:

$$c_{bm} \leq 0.6 \tag{14}$$

for designs with 8 timepoints. Higher values risk non-PD matrices.

Part IV
Condition Number Analysis

12 Definition and Interpretation

The condition number measures numerical stability:

$$\kappa = \frac{\lambda_{max}}{\lambda_{min}} \tag{15}$$

Table 3: Condition Number Interpretation

κ **Value**	**Status**	**Interpretation**
$\kappa < 10$	Excellent	Near-perfect numerical stability
$10 < \kappa < 100$	Good	Acceptable for most applications
$100 < \kappa < 1000$	Poor	Numerical precision concerns
$\kappa > 1000$	Severe	Cholesky decomposition unreliable

Table 4: AR(1) Correlation Decay

Time Gap	**AR(1) Correlation**
6 weeks	$0.75^6 \approx 0.178$
12 weeks	$0.75^{12} \approx 0.032$
18 weeks	$0.75^{18} \approx 0.0056$

13 The AR(1) Problem

AR(1) correlation structure: $\rho_{ij} = \rho^{|t_i - t_j|}$ with $\rho = 0.75$

For evenly-spaced designs (weeks 2, 8, 14, 20), time gaps are 6, 12, 18 weeks:

Correlation ratio: $0.178/0.0056 \approx 32 : 1$

This extreme ratio creates condition numbers $\kappa \approx 600$–$14,621$ for evenly-spaced designs.

14 Why Clustering Solves This

Clustered designs (weeks 4, 8, 9, 10, 11, 12, 16, 20) have mixed gaps:

- Short gaps: 1–4 weeks → correlations 0.32–0.75
- Long gaps: 8–16 weeks → correlations 0.10–0.32

This creates **intermediate correlation values** spanning the correlation space more evenly, producing balanced eigenvalue distributions with $\kappa \approx 56$–61.

15 Alternative Correlation Structures

Five structures are implemented for flexible design:

Table 5: Correlation Structure Comparison

Structure	**Formula**	**Best For**
AR(1)	$\rho^{|t_i - t_j|}$	Clustered designs
Exponential	$\exp(-\lambda|t_i - t_j|)$	Evenly-spaced ($\kappa \approx 18$)
Power Law	$(1 + \alpha|t_i - t_j|)^{-\beta}$	Evenly-spaced ($\kappa \approx 13$)
Matérn	$(1 + \sqrt{3}d/\rho)\exp(-\sqrt{3}d/\rho)$	Flexible smoothness
Rational Quadratic	$(1 + t^2/(2\alpha\lambda^2))^{-\alpha}$	Best stability ($\kappa \approx 10$)

Part V
Three-Factor Response Model

16 Response Decomposition

Each participant's response is decomposed into three components:

$$\text{Response}_t = \text{Baseline} + \text{BR}_t + \text{ER}_t + \text{TR}_t \tag{16}$$

Table 6: Response Component Definitions

Component	**Name**	**Meaning**	**Driver**
BR	Biological Response	True pharmacological effect	Treatment, biomarker
ER	Expectancy Response	Placebo/expectancy effect	Blinding status
TR	Time-variant Response	Natural disease progression	Time, individual

17 Component Mean Formulas

17.1 Biological Response (BR)

$$\text{BR}_{mean,t} = \begin{cases} w_t \cdot r_{eff} & \text{if on treatment} \\ w_{accum} \cdot \gamma \cdot f(t_{off}) & \text{if off treatment (carryover)} \end{cases} \tag{17}$$

where:

- w_t = weeks on drug up to timepoint t
- $r_{eff} = r_0(1 + \beta \cdot \text{BiomarkerCentered})$ = effective BR rate
- γ = carryover proportion
- $f(t_{off})$ = decay function (exponential, power law, etc.)

17.2 Expectancy Response (ER)

$$\text{ER}_{mean,t} = e_t \cdot r_{exp} \tag{18}$$

where:

- $e_t \in \{0, 0.5, 1\}$ = expectancy factor (design-dependent)
- r_{exp} = expectancy response rate

17.3 Time-variant Response (TR)

$$\text{TR}_{mean,t} = f_{time}(t) \tag{19}$$

representing natural disease trajectory independent of treatment.

Table 7: Expectancy Allocation by Design

Design	**Weeks with** $e = 1.0$	**Rationale**
OL+BDC	Weeks ≤ 16	Open-label phase
Hybrid	Weeks 4, 8 only	Unblinded baseline only
Crossover	Varies by period	Blinding maintained

18 Design-Specific Expectancy Patterns

Part VI
Biomarker×Treatment Interaction

19 Two-Level Mechanism

The biomarker-treatment interaction operates at two levels:

19.1 Level 1: Covariance Structure (c_{bm})

The correlation between biomarker and BR components:

$$\text{Corr}(\text{biomarker}, \text{BR}_t) = c_{bm} \tag{20}$$

This creates covariance-level association: participants with high biomarker values tend to have correlated BR values.

19.2 Level 2: Mean Modulation (biomarker_moderation)

The biomarker modifies treatment effect magnitude:

$$r_{eff} = r_0(1 + \beta \cdot \text{BiomarkerCentered}) \tag{21}$$

where β = `biomarker_moderation` parameter.

20 Mathematical Relationship

Table 8: Comparison of Biomarker Effect Mechanisms

Aspect	c_{bm}	**biomarker_moderation**
Level	Covariance	Mean
Effect	Correlation in draws	Scaling of treatment effect
Parameter range	0–0.6 (PD constraint)	0–0.65 (power analysis)
Impact	Noise structure	Signal magnitude

21 Detectability

The biomarker×treatment interaction is tested via:

$$H_0 : \beta_{biomarker \times treatment} = 0 \tag{22}$$

in the mixed model:

```
lmer(response ~ biomarker * treatment + week + (1|participant_id))
```

Power depends on both c_{bm} and biomarker_moderation values.

Part VII
Carryover Effect Modeling

22 The Fundamental Question

Should carryover effects modify correlation parameters in addition to means?

Proposition 1. Carryover affects means only, not correlations.

23 Mathematical Argument

Approach 1 (Incorrect): Carryover modifies correlations

$$\rho_{BR}(t_{on}, t_{off}) = \rho_0 \cdot g(\text{carryover}) \tag{23}$$

This would mean participants' BR correlations change based on their treatment history—biologically implausible for most drugs.

Approach 2 (Correct): Carryover affects means only

$$\mu_{BR}(t_{off}) = \mu_{BR}(t_{on}) \cdot h(t_{since\_discontinuation}) \tag{24}$$

The residual drug effect enters through the expected value, not the variance structure.

24 Double-Counting Problem

If carryover modified both means and correlations:

1. Mean effect: BR_{mean} higher during off-period
2. Correlation effect: BR draws more correlated during off-period

This would count the same biological phenomenon twice, inflating carryover impact and biasing estimates.

Table 9: Carryover Model Comparison

Model	**Formula**	**Parameters**	**Best For**
Fixed Proportion	γ (constant)	1	Simple drugs
Exponential	$\exp(-\lambda t)$	2	Psychiatric drugs
Power Law	$(1+\alpha t)^{-\beta}$	3	Tissue accumulation
Two-Compartment	$Ae^{-\alpha t} + Be^{-\beta t}$	4	Biphasic kinetics
Delayed Elimination	Lag + exponential	3	Prodrugs

25 Five Carryover Models

26 Half-Life Relationship

For exponential decay:

$$t_{1/2} = \frac{\ln(2)}{\lambda} \approx \frac{0.693}{\lambda} \tag{25}$$

Table 10: Half-Life Examples

λ (**week**$^{-1}$)	$t_{1/2}$ (**weeks**)	**Application**
0.05	13.9	Slow-elimination drugs
0.10	6.9	Typical psychiatric medications
0.15	4.6	Faster clearance

Part VIII
Mixed-Effects Model Specification

27 Core Model Formula

$$\begin{aligned}\text{Response}_{it} = \beta_0 + \beta_1 \cdot \text{Biomarker}_i + \beta_2 \cdot \text{Treatment}_{it} \\ + \beta_3 \cdot \text{Biomarker}_i \times \text{Treatment}_{it} + \beta_4 \cdot \text{Week}_{it} + u_i + \epsilon_{it}\end{aligned} \tag{26}$$

where:

- β_3 = biomarker×treatment interaction (primary endpoint)
- $u_i \sim N(0, \sigma_u^2)$ = random intercept
- $\epsilon_{it} \sim N(0, \sigma^2)$ = residual error

28 Model Variants

With Carryover Term:

```
lmer(response ~ biomarker * treatment + week + carryover_effect +
    (1|participant_id))
```

Without Carryover (Hendrickson Original):

```
lmer(response ~ biomarker * treatment + week + (1|participant_id))
```

29 Random Effects Options

Table 11: Random Effects Structures

Structure	Formula	Interpretation
Random intercept	`(1|participant_id)`	Individual baseline differences
Random slope (drug)	`(1+treatment|participant_id)`	Individual treatment response
Random slope (time)	`(1+week|participant_id)`	Individual trajectories

Part IX
Design Comparison Framework

30 Five Trial Designs

Table 12: Trial Design Comparison

Design	Points	Schedule	Carryover
Hybrid	8	Clustered	Moderate-High
OL+BDC	8	Clustered	High
Open-Label	4	Evenly-spaced	None
Crossover	4	Evenly-spaced	Critical
Parallel	4	Evenly-spaced	None

31 Measurement Schedules

31.1 Clustered (Hybrid)

Weeks: $\{4, 8, 9, 10, 11, 12, 16, 20\}$

- Dense cluster at transition (weeks 9–12)
- Captures carryover decay dynamics
- AR(1) works well ($\kappa \approx 56$–61)

31.2 Clustered (OL+BDC)

Weeks: $\{4, 8, 12, 16, 17, 18, 19, 20\}$

- Dense cluster at discontinuation (weeks 16–20)
- Extended washout observation
- AR(1) works well ($\kappa \approx 60$)

31.3 Evenly-Spaced

Weeks: $\{2, 8, 14, 20\}$

- Uniform 6-week gaps
- Requires alternative correlation structures
- Exponential or Power Law recommended ($\kappa \approx 13$–18)

32 Expectancy Vulnerability Analysis

OL+BDC: MORE vulnerable to expectancy confounding

- High expectancy ($e = 1.0$) for 80% of trial (weeks 0–16)
- ER and BR effects coupled at randomization point
- Sudden drop in both components at week 16

Hybrid: LESS vulnerable

- High expectancy only for 20% of trial (weeks 4–8)
- Most measurements during blinded phase ($e = 0.5$)
- Cleaner separation of BR and ER effects

Part X
Statistical Power Analysis

33 Power Definition

$$\text{Power} = P(\text{Reject } H_0 | H_1 \text{ true}) \tag{27}$$

Estimated via Monte Carlo simulation:

$$\hat{\text{Power}} = \frac{\#(\text{iterations with } p < 0.05)}{N_{iterations}} \tag{28}$$

Table 13: Parameter Impact Ranking

Rank	Parameter	Impact Mechanism
1	Design	Measurement schedule, carryover modeling
2	biomarker_moderation	Signal magnitude
3	carryover_t1half	Signal clarity during off-periods
4	c_{bm}	Biomarker-response correlation
5	n_participants	Statistical precision
6	Correlation structure	Numerical stability

Table 14: Power Loss from Ignoring Carryover

Carryover	Without Term	With Term	Type I Error
$\gamma = 0$	∼0.50	∼0.50	0.05
$\gamma = 0.5$	∼0.35	∼0.45	0.07–0.10 (without)
$\gamma = 1.0$	∼0.25	∼0.35	0.10–0.12 (without)

34 Power Determinants (Ranked by Impact)

35 Carryover Cost Analysis

Ignoring carryover leads to:

- 10–20% power reduction
- Inflated Type I error
- Biased treatment effect estimates

Part XI

Alternative Summary Statistics for Interaction Testing

36 Motivation

The primary analysis in this simulation framework uses mixed-effects models to test the biomarker×treatment interaction:

```
lmer(response ~ biomarker * treatment + week + (1|participant_id))
```

While this approach is flexible and accounts for repeated measures, the test statistic (Wald z or t) lacks a simple closed-form power expression. This section develops **alternative summary statistics** that:

1. Permit closed-form power calculations
2. Provide intuition for optimal trial design

3. May sacrifice some efficiency for analytical tractability

37 The Correlation-Based Test Statistic

37.1 Core Intuition

The biomarker×treatment interaction implies that participants with higher biomarker values respond more strongly to treatment. This manifests as a correlation between:

- **Biomarker level**: X_i (baseline covariate)
- **Treatment response differential**: $\Delta Y_i = \bar{Y}_{i,on} - \bar{Y}_{i,off}$

Under the null hypothesis of no interaction, this correlation should be zero. Under the alternative, it should be positive (assuming higher biomarker implies stronger treatment effect).

37.2 Definition

Define the test statistic:

$$\boxed{r = \mathrm{Corr}(X_i, \Delta Y_i)} \tag{29}$$

where:

- X_i = centered biomarker value for participant i
- $\bar{Y}_{i,on}$ = mean response when participant i is on treatment
- $\bar{Y}_{i,off}$ = mean response when participant i is off treatment
- $\Delta Y_i = \bar{Y}_{i,on} - \bar{Y}_{i,off}$ = within-participant treatment effect

37.3 Computation

For each participant $i = 1, \ldots, n$:

$$\bar{Y}_{i,on} = \frac{1}{n_{on,i}} \sum_{t:T_{it}=1} Y_{it} \tag{30}$$

$$\bar{Y}_{i,off} = \frac{1}{n_{off,i}} \sum_{t:T_{it}=0} Y_{it} \tag{31}$$

$$\Delta Y_i = \bar{Y}_{i,on} - \bar{Y}_{i,off} \tag{32}$$

Then compute the sample correlation:

$$r = \frac{\sum_{i=1}^{n}(X_i - \bar{X})(\Delta Y_i - \overline{\Delta Y})}{\sqrt{\sum_{i=1}^{n}(X_i - \bar{X})^2 \sum_{i=1}^{n}(\Delta Y_i - \overline{\Delta Y})^2}} \tag{33}$$

38 Distribution Under the Null

38.1 Fisher's z-Transformation

Under $H_0 : \rho = 0$ (no biomarker×treatment interaction):

$$z = \frac{1}{2} \ln \left(\frac{1+r}{1-r} \right) \approx \mathcal{N} \left(0, \frac{1}{n-3} \right) \tag{34}$$

This transformation stabilizes the variance, making it approximately independent of the true correlation.

38.2 Test Procedure

Reject H_0 at level α if:

$$|z| > z_{1-\alpha/2}/\sqrt{n-3} \tag{35}$$

or equivalently, if:

$$|r| > \frac{z_{1-\alpha/2}}{\sqrt{n-3+z_{1-\alpha/2}^2}} \tag{36}$$

For $\alpha = 0.05$ and $n = 70$:

$$|r| > \frac{1.96}{\sqrt{67+3.84}} \approx 0.233 \tag{37}$$

39 Distribution Under the Alternative

39.1 True Correlation

Under H_1, the true correlation ρ between biomarker and treatment differential depends on:

$$\rho = \frac{\text{Cov}(X_i, \Delta Y_i)}{\sqrt{\text{Var}(X_i) \cdot \text{Var}(\Delta Y_i)}} \tag{38}$$

39.2 Deriving Cov$(X_i, \Delta Y_i)$

Under the linear interaction model:

$$E[\Delta Y_i | X_i] = \tau + \beta \cdot X_i \tag{39}$$

where β is the biomarker×treatment interaction coefficient. Therefore:

$$\text{Cov}(X_i, \Delta Y_i) = \beta \cdot \text{Var}(X_i) \tag{40}$$

39.3 Deriving Var(ΔY_i)

The variance of the treatment differential has two sources:

$$\text{Var}(\Delta Y_i) = \beta^2 \cdot \text{Var}(X_i) + \text{Var}(\epsilon_{\Delta,i}) \tag{41}$$

$$= \beta^2 \sigma_X^2 + \sigma_\Delta^2 \tag{42}$$

where σ_Δ^2 is the residual variance of the treatment effect (reflecting measurement error, within-person variability, etc.).

39.4 Closed-Form Expression for ρ

Combining these:

$$\boxed{\rho = \frac{\beta\sigma_X}{\sqrt{\beta^2\sigma_X^2 + \sigma_\Delta^2}} = \frac{1}{\sqrt{1 + \frac{\sigma_\Delta^2}{\beta^2\sigma_X^2}}}} \tag{43}$$

This has a signal-to-noise interpretation:

$$\rho = \frac{1}{\sqrt{1 + 1/\mathrm{SNR}^2}} \quad \text{where} \quad \mathrm{SNR} = \frac{\beta\sigma_X}{\sigma_\Delta} \tag{44}$$

40 Closed-Form Power Expression

40.1 Power via Fisher's z

Under the alternative with true correlation ρ:

$$z \sim \mathcal{N}\left(\zeta, \frac{1}{n-3}\right) \tag{45}$$

where $\zeta = \frac{1}{2}\ln\left(\frac{1+\rho}{1-\rho}\right)$ is the population Fisher's z.

40.2 Power Formula

Theorem 3 (Power for Correlation Test)**.** The power to detect a biomarker×treatment interaction with true correlation ρ using n participants at significance level α is:

$$\boxed{1 - \beta = \Phi\left(\zeta\sqrt{n-3} - z_{1-\alpha/2}\right) + \Phi\left(-\zeta\sqrt{n-3} - z_{1-\alpha/2}\right)} \tag{46}$$

For one-sided tests (expected positive correlation):

$$1 - \beta = \Phi\left(\zeta\sqrt{n-3} - z_{1-\alpha}\right) \tag{47}$$

40.3 Sample Size Formula

Inverting the power formula for required sample size:

$$\boxed{n = 3 + \left(\frac{z_{1-\alpha/2} + z_{1-\beta}}{\zeta}\right)^2} \tag{48}$$

41 Relationship to Interaction Coefficient

41.1 Converting Between ρ and β

The regression coefficient β (biomarker_moderation) and correlation ρ are related by:

$$\beta = \rho \cdot \frac{\sigma_\Delta}{\sigma_X} \tag{49}$$

or equivalently:

$$\rho = \beta \cdot \frac{\sigma_X}{\sigma_\Delta} \tag{50}$$

Table 15: Required Sample Size for 80% Power ($\alpha = 0.05$, two-sided)

True ρ	**Fisher's ζ**	**Required n**
0.20	0.203	197
0.30	0.310	88
0.40	0.424	49
0.50	0.549	32
0.60	0.693	23

41.2 Interpretation

- β measures the **unstandardized effect**: how many units the treatment differential changes per unit increase in biomarker
- ρ measures the **standardized association**: the correlation between biomarker and treatment response

For power calculations, ρ is more useful because its distribution is known (via Fisher's z).

42 Design Implications

42.1 What Determines ρ?

From the SNR formula, larger ρ (easier detection) occurs when:

1. **Larger** β: Stronger true interaction effect
2. **Larger** σ_X: More biomarker heterogeneity in sample
3. **Smaller** σ_Δ: Less noise in treatment differential

42.2 Reducing σ_Δ

The variance of the treatment differential can be decomposed:

$$\sigma_\Delta^2 = \frac{\sigma_{on}^2}{n_{on}} + \frac{\sigma_{off}^2}{n_{off}} + 2\sigma_{\text{within}}^2 \tag{51}$$

where:

- $\sigma_{on}^2, \sigma_{off}^2$ = measurement error variance
- n_{on}, n_{off} = number of measurements in each condition
- σ_{within}^2 = within-person variability

Design implication: More measurements per participant reduce σ_Δ^2, increasing ρ and power.

42.3 Balance Between Conditions

For fixed total measurements $n_{on} + n_{off} = m$, σ^2_Δ is minimized when $n_{on} = n_{off} = m/2$ (balanced allocation).

Design implication: Balanced designs (equal on/off measurements) maximize power for detecting biomarker×treatment interactions.

42.4 Recruitment Strategy

Power increases with σ_X (biomarker variance). This suggests:

1. Recruit from heterogeneous populations
2. Avoid restricting biomarker range
3. Consider stratified sampling to ensure biomarker spread

43 Comparison to Mixed-Effects Approach

Table 16: Comparison of Interaction Test Approaches

Aspect	**Correlation Test**	**Mixed Model**
Test statistic	r (or Fisher's z)	Wald z or t
Power formula	Closed-form (Fisher)	Monte Carlo only
Repeated measures	Collapsed to ΔY	Modeled explicitly
Covariates	Not directly included	Easily added
Efficiency	Lower (loses information)	Higher (uses all data)
Robustness	Less model-dependent	Assumes linearity, normality

43.1 When to Use Each

Correlation test:

- Power/sample size planning (closed-form calculations)
- Quick exploratory analysis
- When model assumptions are uncertain

Mixed model:

- Primary confirmatory analysis
- When covariates must be controlled
- When repeated measures structure is important

44 Extension: Slope-Based Statistics

44.1 Regression of ΔY on X

An alternative to correlation is the regression slope:

$$\Delta Y_i = \alpha + \beta X_i + \epsilon_i \tag{52}$$

The OLS estimate:

$$\hat{\beta} = \frac{\sum_i (X_i - \bar{X})(\Delta Y_i - \overline{\Delta Y})}{\sum_i (X_i - \bar{X})^2} \tag{53}$$

Under standard assumptions:

$$\hat{\beta} \sim \mathcal{N}\left(\beta, \frac{\sigma_\Delta^2}{\sum_i (X_i - \bar{X})^2}\right) \tag{54}$$

44.2 Power for Slope Test

$$1 - \beta = \Phi\left(\frac{|\beta|\sqrt{\sum (X_i - \bar{X})^2}}{\sigma_\Delta} - z_{1-\alpha/2}\right) \tag{55}$$

This is equivalent to the correlation test but expressed in terms of the regression coefficient rather than the correlation.

45 Summary of Key Formulas

45.1 Test Statistic

$$r = \text{Corr}(X_i, \Delta Y_i) = \text{Corr}(\text{Biomarker}, \bar{Y}_{on} - \bar{Y}_{off}) \tag{56}$$

45.2 Signal-to-Noise Ratio

$$\rho = \frac{1}{\sqrt{1 + \sigma_\Delta^2 / (\beta^2 \sigma_X^2)}} \tag{57}$$

45.3 Power

$$1 - \beta = \Phi\left(\zeta\sqrt{n-3} - z_{1-\alpha/2}\right) \tag{58}$$

45.4 Sample Size

$$n = 3 + \left(\frac{z_{1-\alpha/2} + z_{1-\beta}}{\zeta}\right)^2 \tag{59}$$

45.5 Design Principle

To maximize power for detecting biomarker×treatment interactions: maximize biomarker heterogeneity (σ_X), increase measurements per condition (n_{on}, n_{off}), and maintain balanced allocation.

46 Literature Review: Analytical Power Alternatives

This section surveys analytical approaches to power calculation in settings comparable to Hendrickson et al. (2020), where mixed-effects models test biomarker×treatment interactions in N-of-1 or crossover designs.

46.1 Why Analytical Power Is Difficult

The mixed-effects model

```
lmer(response ~ biomarker * treatment + week + (1|participant_id))
```

involves a Wald test statistic whose exact distribution depends on:

1. The design matrix structure X
2. The variance-covariance matrix V
3. The variance components (σ_u^2, σ^2)

Computing $\text{Var}(\hat{\beta}) = \sigma^2(X'V^{-1}X)^{-1}$ requires inverting a large block-diagonal matrix whose structure varies by design. For Hendrickson's clustered measurement schedules with AR(1) correlation, this inversion lacks a simple closed form.

46.2 Summary Measures Analysis

The most widely used analytical alternative collapses repeated measures to participant-level summaries.

Key references:

- Matthews, Altman, Campbell, Royston (1990) – "Analysis of serial measurements in medical research" (BMJ)
- Frison & Pocock (1992) – "Repeated measures in clinical trials: analysis using mean summary statistics"

Approach: Compute a summary statistic per participant (mean, AUC, slope, change score), then apply standard between-subject tests. The correlation $r = \text{Corr}(X_i, \Delta Y_i)$ developed in this document is a summary measure approach.

Trade-off: Loses efficiency (ignores within-person correlation structure) but gains analytical tractability. Efficiency loss is moderate when intraclass correlation (ICC) is high (0.5–0.8), which is typical in N-of-1 settings.

46.3 The "Rule of $4\times$" for Interactions

Key reference: Brookes et al. (2004) – "Subgroup analyses in randomised trials: risks of subgroup-specific analyses" (Statistics in Medicine)

Key finding: Detecting a treatment×covariate interaction requires approximately **4 times the sample size** compared to detecting the main treatment effect of the same magnitude.

Mathematical basis:

$$\text{Var}(\hat{\beta}_{interaction}) \approx 4 \cdot \text{Var}(\hat{\beta}_{treatment}) \tag{60}$$

This arises because the interaction term has reduced "effective sample size"—only the covariance between biomarker and treatment contributes to the signal.

Implication: If detecting a main treatment effect requires $n = 20$, detecting the same-sized interaction requires $n \approx 80$.

46.4 Asymptotic Mixed Model Formulas

Key references:

- Shieh (2003) – "On power and sample size calculations for likelihood ratio tests in generalized linear models"
- Stroup (2002) – "Power analysis based on spatial effects mixed models"
- Demidenko (2004) – *Mixed Models: Theory and Applications* (Chapter 10)

Approach: Use the noncentrality parameter of the Wald or likelihood ratio test:

$$\text{ncp} = \frac{\beta^2}{\text{Var}(\hat{\beta})} = \frac{\beta^2}{\sigma^2 (X'V^{-1}X)^{-1}_{jj}} \tag{61}$$

Then:

$$\text{Power} = \Phi\left(\frac{|\beta|}{\text{SE}(\hat{\beta})} - z_{1-\alpha/2}\right) \tag{62}$$

Challenge: Computing $(X'V^{-1}X)^{-1}$ requires specifying the design matrix, variance components, and correlation structure. For balanced designs with compound symmetry, closed forms exist. For Hendrickson's clustered designs with AR(1), the algebra becomes intractable.

46.5 GEE-Based Power

Key references:

- Liu & Liang (1997) – "Sample size calculations for studies with correlated observations"
- Rochon (1998) – "Application of GEE procedures for sample size calculations in repeated measures experiments"

Formula for treatment effect:

$$n = \frac{2(z_{\alpha/2} + z_\beta)^2 \sigma^2}{\delta^2} \times \text{Design Effect} \tag{63}$$

where Design Effect $= \frac{1+(m-1)\rho}{m}$ for compound symmetry with m measurements and correlation ρ.

Limitation: Primarily developed for main effects, not treatment×covariate interactions.

46.6 Crossover Trial Formulas

Key references:

- Jones & Kenward (2014) – *Design and Analysis of Cross-Over Trials* (3rd edition)
- Senn (2002) – *Cross-over Trials in Clinical Research* (2nd edition)

For standard 2×2 crossover (treatment effect):

$$n = \frac{2(z_{\alpha/2} + z_\beta)^2 \sigma_d^2}{\delta^2} \tag{64}$$

where σ_d^2 is the within-subject variance of differences.
For treatment$\times$period interaction (analogous to treatment$\times$biomarker):

$$n_{interaction} \approx 4 \times n_{treatment} \tag{65}$$

Limitation: These formulas assume balanced designs. Hendrickson's Hybrid and OL+BDC designs are unbalanced (different numbers of on/off measurements).

46.7 Design Effect Approach

Key reference: Diggle, Heagerty, Liang, Zeger (2002) – *Analysis of Longitudinal Data* (Chapter 13)
Approach: Express power in terms of a "design effect" that inflates variance:

$$\text{Var}(\hat{\beta}) = \text{Var}_{simple} \times DE \tag{66}$$

For AR(1) correlation with m equally-spaced timepoints:

$$DE = \frac{1}{m}\left[1 + 2\sum_{k=1}^{m-1}\left(1 - \frac{k}{m}\right)\rho^k\right] \tag{67}$$

For Hendrickson's clustered designs: The unequal spacing (weeks 4, 8, 9, 10, 11, 12, 16, 20) requires computing:

$$DE = \frac{1}{m^2}\sum_{i,j}\rho^{|t_i - t_j|} \tag{68}$$

This can be computed numerically but does not yield a simple closed form.

46.8 Bayesian Assurance

Key reference: O'Hagan & Stevens (2001) – "Bayesian assessment of sample size for clinical trials of cost-effectiveness"
Approach: Instead of frequentist power (probability of significance given fixed true effect), compute "assurance" (probability of significance averaging over prior uncertainty about true effect):

$$\text{Assurance} = \int \text{Power}(\theta) \cdot \pi(\theta)\, d\theta \tag{69}$$

Advantage: More realistic when effect size is uncertain.
Disadvantage: Requires specifying a prior; still often computed by simulation.

Table 17: Comparison of Power Calculation Methods

Approach	**Analytical?**	**Interaction?**	**Unbalanced?**	**Accuracy**
Summary measures	Yes	Yes	Partial	Moderate
Rule of 4×	Yes	Yes	No	Approximate
Asymptotic mixed	Semi	Yes	Numerical	Good
GEE-based	Yes	Limited	Partial	Moderate
Crossover formulas	Yes	Limited	No	Good (balanced)
Design effect	Semi	Limited	Yes	Moderate
Simulation	No	Yes	Yes	Excellent

46.9 Comparison of Analytical Approaches

46.10 The Efficiency-Tractability Trade-off

Three approaches represent different points on the efficiency-tractability spectrum:

1. **Full mixed model** (Hendrickson approach): Uses all data optimally but requires simulation for power.

2. **Summary correlation** (this document): Loses ∼15–30% efficiency but has closed-form power via Fisher's z.

3. **Rule of** $4\times$: Very rough approximation—if main effect needs n, interaction needs $4n$.

The summary approach is well-established in the literature (Matthews et al., 1990). It is particularly appropriate when ICC is high (typical in N-of-1: 0.5–0.8), where collapsing measurements loses minimal information.

46.11 Recommended Analytical Approximation

For Hendrickson-style power calculations, a practical analytical formula combines:

Step 1: Estimate effective correlation from design parameters

$$\rho_{eff} = \frac{\beta \cdot \sigma_X}{\sqrt{\beta^2 \sigma_X^2 + \sigma_\Delta^2}} \tag{70}$$

where $\sigma_\Delta^2 \approx \frac{\sigma^2}{n_{on}} + \frac{\sigma^2}{n_{off}}$ (measurement noise reduced by averaging).

Step 2: Apply Fisher's z power formula

$$n = 3 + \left(\frac{z_{1-\alpha/2} + z_{1-\beta}}{\zeta} \right)^2 \tag{71}$$

Step 3: Apply design-specific inflation factor

This hybrid approach provides analytical tractability while acknowledging design-specific features through calibrated inflation factors.

Table 18: Design-Specific Inflation Factors

Design	**Inflation**	**Rationale**
Hybrid	$\times 1.1$	Near-balanced on/off measurements
OL+BDC	$\times 1.2$	Unbalanced (more on than off)
Crossover	$\times 1.1$	Balanced but carryover concerns
Parallel	$\times 1.5$	Between-subject comparison loses power

46.12 Key References Summary

1. Matthews et al. (1990) – Summary measures analysis (BMJ)
2. Frison & Pocock (1992) – Mean summary statistics
3. Brookes et al. (2004) – Rule of 4× for interactions
4. Shieh (2003) – Asymptotic mixed model power
5. Liu & Liang (1997) – GEE-based power
6. Jones & Kenward (2014) – Crossover trial design
7. Senn (2002) – Cross-over trials in clinical research
8. Diggle et al. (2002) – Analysis of longitudinal data
9. Demidenko (2004) – Mixed models theory
10. O'Hagan & Stevens (2001) – Bayesian assurance

Part XII
Key Mathematical Insights

47 The Eigenvalue-Conditioning Nexus

The condition number problem is fundamentally about eigenvalue balance:

$$\kappa = \frac{\lambda_{max}}{\lambda_{min}} \tag{72}$$

Good conditioning requires eigenvalues to be similar in magnitude, which requires correlation values to span the correlation space evenly rather than clustering at extreme values.

48 The Clustering Insight

Clustered measurement schedules succeed because they create **intermediate correlations**:

- AR(1) with 1-week gaps: $\rho = 0.75$
- AR(1) with 4-week gaps: $\rho = 0.32$

- AR(1) with 8-week gaps: $\rho = 0.10$

This natural variation from clustering balances eigenvalues.

Evenly-spaced designs with uniform large gaps (6 weeks) create extreme correlation ratios (32:1), leading to extreme eigenvalue ratios.

49 The Carryover-Correlation Separation

The principle that carryover affects means only (not correlations) is essential for:

1. **Avoiding double-counting** of carryover effects
2. **Maintaining interpretability** of correlation parameters
3. **Preserving positive definiteness**

50 The Hierarchy Constraint

The correlation hierarchy ($c_{cfct} < c_{cf1t} < c_{autocorr}$) encodes the temporal coherence principle:

> *Measurements at different times cannot be more correlated than measurements at the same time.*

Violating this creates mathematically impossible constraint systems.

Appendices

A Key Formulas Reference

A.1 Covariance Matrix

$$\Sigma = D \cdot R \cdot D \tag{73}$$

A.2 Conditional Mean

$$\mu_{1|2} = \mu_1 + \Sigma_{12}\Sigma_{22}^{-1}(x_2 - \mu_2) \tag{74}$$

A.3 Conditional Covariance

$$\Sigma_{cond} = \Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{12}' \tag{75}$$

A.4 Condition Number

$$\kappa = \frac{\lambda_{max}}{\lambda_{min}} \tag{76}$$

A.5 AR(1) Correlation

$$\rho_{ij} = \rho^{|t_i - t_j|} \tag{77}$$

A.6 Exponential Correlation

$$\rho_{ij} = \exp(-\lambda |t_i - t_j|) \tag{78}$$

A.7 Carryover Exponential Decay

$$\text{Carryover}_t = \text{Effect}_0 \cdot \exp\left(-\frac{\ln(2)}{t_{1/2}} \cdot t_{off}\right) \tag{79}$$

A.8 Half-Life Relationship

$$t_{1/2} = \frac{\ln(2)}{\lambda} \tag{80}$$

A.9 Hierarchy Constraint

$$c_{cfct} < c_{cf1t} < \min(c_{tv}, c_{pb}, c_{br}) \tag{81}$$

B Document Sources

This synthesis draws from the following documentation files:

1. `sigma_matrix_derivation.tex` – Core $\Sigma = DRD$ derivation
2. `positive_definiteness_constraints.tex` – PD analysis
3. `correlation_structure_design.tex` – Block structure design
4. `carryover_correlation_theory.tex` – Carryover separation
5. `correlation_structure_white_paper.Rmd` – AR(1) alternatives
6. `carryover_modeling_white_paper.Rmd` – Five carryover models
7. `mixed_model_comparison_and_best_practices.md` – Model guidance
8. `technical_differences_scaling_and_pd.tex` – Implementation
9. `correlation_parameters_guide.tex` – Parameter interpretation
10. `whitepaper.tex` – Crossover vs N-of-1 comparison
11. Part XI (Alternative Summary Statistics) – Original theoretical development using Fisher's z-transformation for closed-form power analysis

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